**Tribasic Acid:** An acid which gives three hydrogen ions per molecule of the acid in solution. Example: $H_3PO_4$

**(c) Based on Ionisation**

Acids get ionised in water (produce $H^+$ ions) completely or partially. Based on the extent of ionisation acids are classified as below.

**Strong Acids:** These are acids that ionise completely in water. Example: HCl

**Weak Acids:** These are acids that ionise partially in water. Example: $CH_3COOH$.




Ionisation is the condition of being dissociated into ions by heat or radiation or chemical reactions or electrical discharge.

**(d) Based on Concentration**

**Concentrated Acid:** It has relatively large amount of acid dissolved in a solvent.

**Dilute Acid:** It has relatively smaller amount of acid dissolved in solvent.

## 14.1.2 Properties of Acids

a) They have sour taste.

b) Their aqueous solutions conduct electricity since they contain ions.

c) Acids turns blue litmus red.

d) Acids react with active metals to give hydrogen gas.

$Mg + H_2SO_4 \rightarrow MgSO_4 + H_2 \uparrow$

$Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$




Few metals do not react with acids.
Example: Ag, Cu.

e) Acids react with metal carbonate and metal hydrogen carbonate to give carbon dioxide.

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2 \uparrow$

$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2 \uparrow$

f) Acids react with metallic oxides to give salt and water.

$CaO + H_2SO_4 \rightarrow CaSO_4 + H_2O$

g) Acids react with bases to give salt and water.

$HCl + NaOH \rightarrow NaCl + H_2O$

The reaction is known as neutralisation reaction.

### Activity 1

Take about 10 ml of dilute hydrochloric acid in a test tube and add a few pieces of zinc granules into it. What do you observe? Why are bubbles formed in the solution? Take a burning candle near a bubble containing hydrogen gas, the flame goes off with a 'Popping' sound. This confirms that metal displaces hydrogen gas from the dilute acid.

**Caution:** Care must be taken while mixing any concentrated inorganic acid with water. The acid must be added slowly and carefully with constant stirring to water since it generates large amount of heat. If water is added to acid, the mixture splashes out of the container and it may cause burns.

## 14.1.3 Uses of Acids

- Sulphuric acid is called King of Chemicals because it is used in the preparation of many other compounds. It is used in car batteries also.
- Hydrochloric acid is used as a cleansing agent in toilets.
- Citric acid is used in the preparation of effervescent salts and as a food preservative.
- Nitric acid is used in the manufacture of fertilizers, dyes, paints and drugs.
- Oxalic acid is used to clean iron and manganese deposits from quartz crystals. It is also used as bleach for wood and removing black stains.
- Carbonic acid is used in aerated drinks.
- Tartaric acid is a constituent of baking powder.



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